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SARDAR PATEL UNIVERSITY
BCA Semester-4 Examination
US04CBCA52 || DATA STRUCTURE -II

Date : 7-8-2023

Time : 10.00 a.m. – 1.00 p.m.

Total : 70 Marks

Q-1 Select the appropriate option for the following questions:

10

- 1) _____ contains maximum possible number of nodes in all level.
A. Skew binary tree
B. Full binary tree
C. Complete binary tree
D. None of these.
- 2) Leaf node is known as _____ node.
A. Terminal
B. Root
C. Parent
D. Non-terminal
- 3) Each node in singly linked list has _____ fields.
A. 2
B. 3
C. 1
D. 4
- 4) Which of the following is NOT the type of Linked list?
A. Circular linked list
B. Hybrid linked list
C. Doubly linked list
D. Singly linked list
- 5) _____ is the process to find the particular element from the array.
A. Sorting
B. Inserting
C. Searching
D. Updating
- 6) Binary search algorithms performs efficiently on a ...
A. array
B. linked list
C. Both A & B
D. None of these
- 7) IBM stands for _____.
A. International Business Machine
B. Internet Business Machine
C. International Business manager
D. None of these
- 8) The lowest level of index is _____.
A. Track index
B. Prime index
C. Index area
D. Master index
- 9) In quick sort method, a desired choice for partitioning element will be ...
A. first element of list
B. last element of list
C. any element of list
D. median of list
- 10) _____ means a link between parent and its child.
A. Degree
B. Height.
C. Leaf
D. Branch

Q-2 Give short answers of the following questions: (Attempt TEN out of TWELVE)

20

- 1) List out Applications of Tree.
- 2) Define Following Terms: Degree of Node & Path.
- 3) Explain Binary Search Tree with an example.
- 4) Explain circular linked list in brief.
- 5) Explain doubly linked list in brief.

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- 6) Explain linked list in brief.
- 7) List out any five applications of sorting.
- 8) Define: Internal Sort and External Sort.
- 9) Trace the given data using Bubble sort: 5, 1, 4, 2, 8.
- 10) Define: Hashing function, Collision resolution technique.
- 11) Explain in brief prime area.
- 12) List types of Chaining.

- Q-3 A Define Traversal. Discuss any one traversal with algorithm. 05
 B Explain Linear (Sequential) representation of a binary tree. 05

OR

- Q-3 A Define Lexically ordered binary tree. Write a detail note on : Insertion operation performed on lexically ordered binary tree. 05
 B Draw a binary tree for the expression $A*B-(C+D)*(P/Q)$. 05

- Q-4 A Write an algorithm to insert node at front position in singly linked list. 05
 B Write an algorithm to delete node from end position in singly linked list. 05

OR

- Q-4 A Write an algorithm to insert node at any position in singly linked list. 05
 B Write an algorithm to delete node at beginning in singly linked list. 05

- Q-5 A Write down the algorithm of bubble sort. 06
 B Write down the algorithm of Linear Search. 04

OR

- Q-5 A Differentiate between Sorting and Searching. 06
 B Write down the algorithm of Quick sort. 04

- Q-6 Explain Mid Square Method, Folding Method and Digit Analysis Method with example. [10]

OR

- Q-6 Write a detail note on structure and processing of Direct files. [10]